GIHAN LAKMAL

AI/ML Engineer | Reasercher | SE Engineer | Agentic AI & Multi-Agent Systems | RAG, LLMs & MLOps

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ABOUT

AI/ML Engineer with nearly 2 years of experience designing, building, and deploying agentic AI systems, machine learning, and generative AI solutions. Specialized in building autonomous AI agents, multi agent orchestration (LangChain, LangGraph), tool/function calling, and LLM-driven workflow automation alongside RAG, NLP, computer vision, and time-series forecasting. Experienced with FastAPI, ChromaDB, Neo4j, PostgreSQL, Docker, MLOps, DevOps, CI/CD, monitoring, and performance optimization for scalable, production-grade AI systems and end-to-end client delivery.

EDUCATION

University of Kelaniya - Sri Lanka (July 2022 - July 2026)

B.Sc. (Hons) Computer Science (AI Specialization) - GPA 3.61

University of Colombo - Sri Lanka (July 2021 - August 2022)

Bachelor of Information Technology (External)

Nalanda College, Colombo 10

G.C.E. Advanced Level 2020 - Physical Science Stream (Combined Mathematics, Physics, Chemistry), Z-Score: 1.563

WORK EXPERIENCE

ArcheAI - Co-Founder (Dec 2025 - Present, part-time)

- Co-founded ArcheAI, an AI research and engineering startup building agentic AI systems that combine multi-agent orchestration, autonomous task execution, and scalable software infrastructure for enterprise and client automation, including an agentic WhatsApp bot, fax extraction models, and an agentic voice call agent platform.
- Designed complete agent architectures using LLM orchestration, tool integration, and workflow automation, applying RAG, semantic embedding optimization, and hybrid neuro-symbolic workflows to enhance contextual reasoning and reliability.

AI/ML Trainee Associate Engineer - SenzMate AIoT (July 2025 – July 2026)

- Designed and implemented modular AI agent nodes and autonomous workflow pipelines for the Interplay Autonomous Platform, enabling scalable multi-agent automation, task routing, memory management, and autonomous decision-making for enterprise users.
- Built finance-focused multi-agent systems using LangChain, LangGraph, and modern LLM orchestration frameworks, with custom tool integration for dynamic agent workflows and Retrieval Augmented Generation (RAG) pipelines.
- Delivered real-world client projects with end-to-end ownership spanning agent architecture, AI model integration, backend development, deployment, and production optimization.

AI Research Assistant (November 2024 - June 2025)

- Worked on generative AI models for Sri Lankan native languages (Sinhala and Tamil); collected and processed a 40-million-word Tamil/Sinhala corpus, improving model accuracy and relevance for native speakers.
- Applied generative AI, computer vision, and audio generation techniques, with LLM fine-tuning (PEFT, LoRA, QLoRA) for real-world language model improvements.

PROJECTS

1. VInvest AI Valuation Intelligence Engine - Multi-Agent Financial AI Platform (Client Project)

- Built the AI “brain” behind an AI powered financial valuation platform: autonomous agents interpret a user's query, classify company type, retrieve financial signals, select the right valuation framework, and run valuation logic end to end.
- **Agentic Architecture:** Implemented multi-agent orchestration with supervisor graph architecture, ReAct style agent reasoning, and Deep Research / Deep QA agent workflows using LangGraph, coordinating specialized agents for data retrieval, reasoning, valuation, and automated report generation.

- integrating custom tool calling and function callings for financial data retrieval across PostgreSQL, MongoDB, Neo4j and Redis.
- Owned QA/load testing and deployment on GKE , validated agent workflows via Docker, Swagger, and live API testing across multiple ticker types (AAPL, AMT, JPM, PGR, JOBY).

  Tech Stack: **Python, FastAPI, LangChain, LangGraph, Agent Architectures , Redis , ChromaDB, Neo4j, PostgreSQL, MongoDB, RAG, Agent Orchestration, Tool Calling, MCP , Function callings , Agent Orchestration , Docker, GKE**

2. Advanced Agentic Hybrid OCR Project - Prior Authorization Fax Intelligence (Client Project, ArcheAI)

- Designed and deployed a production-grade agentic OCR document-processing system for autonomous data extraction from complex, multi-page insurance approval fax documents, achieving 94-98% extraction accuracy.
- **Autonomous Task Execution:** Built a hybrid classifier and extraction agent combining regex-based extraction, LLM-based validation/correction, and vision-language model gap-filling, with confidence scoring that routes low-confidence fields to a human-in-the-loop review step.
- Built a 7-phase asynchronous agent pipeline with Temporal, FastAPI, PostgreSQL, and Docker, including job tracking, extraction storage, audit logging, and multi-tenant support, optimized for 100+ documents/hour.

  Tech Stack: **Python, FastAPI, Temporal, PostgreSQL, OpenAI Vision Models, Tesseract/Paddle OCR, React, Docker, Agent Orchestration**

3. AI Medical Appointment Voice Agentic Platform (USA Healthcare Client)

- Built an end-to-end autonomous voice agent handling live medical appointment calls, using LangGraph to route conversations through agent states (greeting, availability check, rescheduling, cancellation, general Q&A, wrap-up).
- Integrated real-time STT (Deepgram), LLM reasoning (Gemini/Groq), a local RAG knowledge base for clinic-specific Q&A, and TTS with voice cloning, optimizing latency via fast-path intent handling and response caching.

  Tech Stack: **LangGraph, FastAPI, WebSocket Streaming, Deepgram STT, Gemini, Groq, RAG, Voice Cloning, Prompt Engineering**

4. Construction Worker Safety Monitoring System *(* YOLOv8 AI Safety Detection MVP for Indian Client)

- Built an MVP computer vision system detecting workers, helmets, and safety vests from images/video using a trained YOLOv8 model, applying compliance rules to classify safe/unsafe status and generate annotated reports.

  Tech Stack: **YOLOv8, Ultralytics, Python, FastAPI, OpenCV, Docker, PostgreSQL**

5. Medical Drug Sales Prediction Based on Specific Areas in Sri Lanka (IEEE Research Publication)

- Built a production-grade time-series forecasting and inference pipeline for weekly, area-level pharmaceutical sales, ensembling GRU/LSTM, Transformer/TFT, N-BEATS, LightGBM/XGBoost, and Prophet models.
- Delivered SHAP-based explainability and causal/counterfactual analysis, with containerized inference (Docker, Flask/FastAPI), CI tests, and drift monitoring.

  Tech Stack: **Python, GRU/LSTM, Transformer/TFT, LightGBM, XGBoost, Prophet, SHAP, Docker, Flask, FastAPI**

SKILLS

Agentic AI & Orchestration: AI Agents, Multi-Agent Systems, LangChain, LangGraph, ReAct Agents, Supervisor Graph Architecture, Tool/Function Calling, Autonomous Task Execution, LangSmith, Temporal , Agent Orchestrations , ,Architecture Building.

AI & Data Science: Machine Learning (Supervised/Unsupervised), Deep Learning, NLP, Generative AI, RAG, Transformer Models, Prompt Engineering, Computer Vision, Time-Series Forecasting

Programming Languages: Python, Java, JavaScript, React , CSS , html , Nextjs ,C, C# ,

Technologies: TensorFlow, Keras, Scikit-learn, PyTorch-style pipelines, Pandas, NumPy, ARIMA, SARIMAX, FastAPI, ChromaDB, Neo4j, PostgreSQL, MongoDB

MLOps / DevOps: Docker, Kubernetes, CI/CD, MLflow, DVC, Prometheus, Grafana, Model Monitoring & Logging

Tools: Git/GitHub, Jenkins, Jupyter, VS Code, Anaconda, GCP

REFERENCES

Arunn Balasundaram
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SenzMate Pvt Ltd | AI Consultant at Iterate.io
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Prof. Nayomal Dias
B.Sc, M.Sc, PhD | Senior Professor,
University of Kelaniya |
Faculty of Computing and Technology